

Worksheet 3

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Subject Name: Internet Of Things

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1. Aim/Overview of the practical:

Interface an Arduino Uno with Ultrasonic Sensor to calculate the distance and print on a Serial monitor and on LCD. Follow the mentioned cases after the calculation of distance.

- If the distance is > 250 cm green light is in ON condition.
- If the distance is < 250 cm and the distance > 90 yellow light is in ON condition and the green light remains Off
- If the distance is < 90 cm turn on the buzzer and lights will remain OFF.

2. Apparatus:

Hardware Requirements: Arduino Uno Board
Ultrasonic Sensor
LCD (16*2)

Software requirements: TinkerCAD
MS Windows 10

3. Circuit Diagram(TinkerCad): (Case 1)

Coding:

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

// Ultrasonic sensor
const int trigPin = 9;
const int echoPin = 10;

// Green LED
const int greenLED = 6;
void setup(){
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(greenLED, OUTPUT);

  Serial.begin(9600);
  lcd.begin(16, 2);
  lcd.print("Distance:");
}
```

void loop()

```
{
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
```

```
long duration = pulseIn(echoPin, HIGH);
float distance = duration * 0.0343 / 2;
Serial.print("Distance: ");
Serial.print(distance);
Serial.println(" cm");
```

```
lcd.setCursor(0, 0);
lcd.print("Distance:   ");
lcd.setCursor(0, 1);
lcd.print(distance);
lcd.print(" cm   ");
```

```
if (distance > 250){
  digitalWrite(greenLED, HIGH);
  Serial.println("GREEN LED ON");
}
else{
  digitalWrite(greenLED, LOW);
}
delay(500);
```

}

OUTPUT:

Anindita_Dhar_25MCA20259

Simulator time: 00:00:38

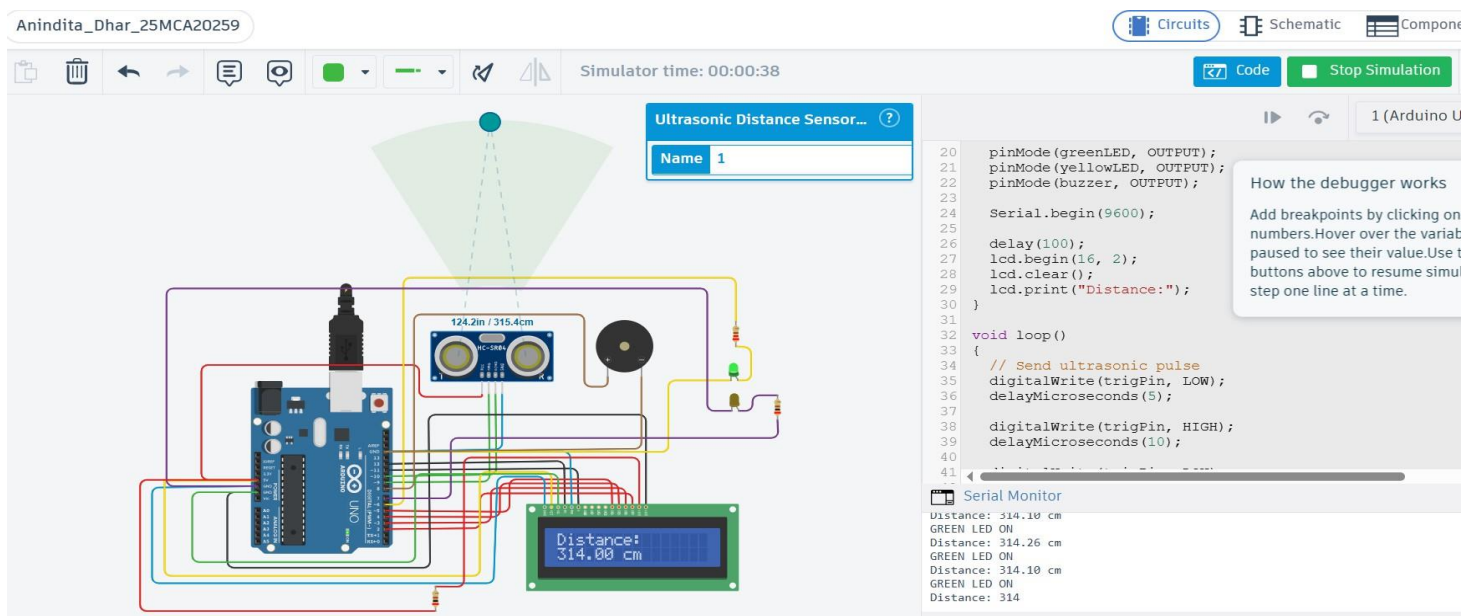
1 (Arduino U

How the debugger works
Add breakpoints by clicking on numbers. Hover over the variable to see their value. Use the buttons above to resume simulation one line at a time.

```
20 pinMode(greenLED, OUTPUT);
21 pinMode(yellowLED, OUTPUT);
22 pinMode(buzzer, OUTPUT);
23
24 Serial.begin(9600);
25
26 delay(100);
27 lcd.begin(16, 2);
28 lcd.clear();
29 lcd.print("Distance:");
30
31
32 void loop()
33 {
34   // Send ultrasonic pulse
35   digitalWrite(trigPin, LOW);
36   delayMicroseconds(5);
37
38   digitalWrite(trigPin, HIGH);
39   delayMicroseconds(10);
40
41 }
```

Serial Monitor

```
Distance: 314.10 cm
GREEN LED ON
Distance: 314.26 cm
GREEN LED ON
Distance: 314.10 cm
GREEN LED ON
Distance: 314
```



(Case 2)**Coding:**

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

// Ultrasonic sensor
const int trigPin = 9;
const int echoPin = 10;
// Yellow LED
const int yellowLED = 7;
void setup()
{
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(yellowLED, OUTPUT);
  Serial.begin(9600);
  lcd.begin(16, 2);
  lcd.print("Distance:");
}

void loop()
{
  // Send ultrasonic pulse
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Measure echo time
  long duration = pulseIn(echoPin, HIGH);

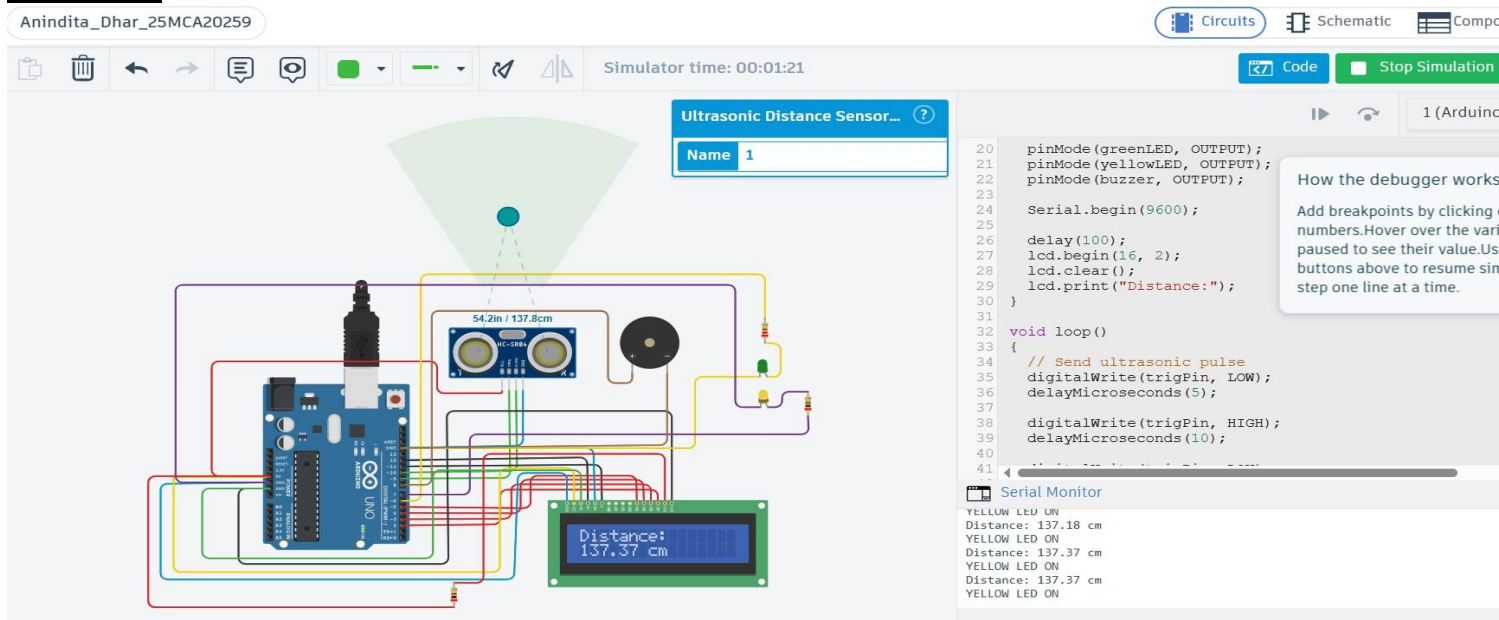
  // Calculate distance in cm
  float distance = duration * 0.0343 / 2;

  // Print on Serial Monitor
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  // Print on LCD
  lcd.setCursor(0, 0);
  lcd.print("Distance:   ");
  lcd.setCursor(0, 1);
  lcd.print(distance);
  lcd.print(" cm   ");
```

```
if (distance < 250 && distance > 90)
{
    digitalWrite(yellowLED, HIGH);
    Serial.println("YELLOW LED ON");
}
else
{
    digitalWrite(yellowLED, LOW);
}
delay(500);
}
```

OUTPUT:



The screenshot shows an Arduino IDE simulation. The circuit includes an Arduino Uno, an ultrasonic sensor (HC-SR04), a yellow LED, a buzzer, and an LCD display (16x2). The ultrasonic sensor is connected to the Arduino's trigPin (9) and echoPin (10). The yellow LED is connected to the Arduino's digital output pins. The buzzer is connected to the Arduino's digital output pins. The LCD display is connected to the Arduino's I2C pins. The code on the right shows the setup and loop functions. The Serial Monitor shows the output: 'YELLOW LED ON', 'Distance: 137.18 cm', 'YELLOW LED ON', 'Distance: 137.37 cm', 'YELLOW LED ON', 'Distance: 137.37 cm', 'YELLOW LED ON'.

(Case 3)

Coding:

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
// Ultrasonic sensor
const int trigPin = 9;
const int echoPin = 10;
// Buzzer
const int buzzer = 8;

void setup()
{
    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
    pinMode(buzzer, OUTPUT);
    Serial.begin(9600);
    lcd.begin(16, 2);
}
```

```

lcd.print("Distance:");
}

void loop()
{
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  long duration = pulseIn(echoPin, HIGH);
  float distance = duration * 0.0343 / 2;
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  lcd.setCursor(0, 0);
  lcd.print("Distance:   ");
  lcd.setCursor(0, 1);
  lcd.print(distance);
  lcd.print(" cm   ");
  if (distance < 90)
  {
    digitalWrite(buzzer, HIGH);
    Serial.println("BUZZER ON");
  }
  else
  {
    digitalWrite(buzzer, LOW);
  }
  delay(500);
}

```

OUTPUT:

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Simulator time: 00:01:40

Ultrasonic Distance Sensor... ?

Name 1

23.7in / 60.1cm

Distance: 60.01 cm

```

20 pinMode(greenLED, OUTPUT);
21 pinMode(yellowLED, OUTPUT);
22 pinMode(buzzer, OUTPUT);
23
24 Serial.begin(9600);
25
26 delay(100);
27 lcd.begin(16, 2);
28 lcd.clear();
29 lcd.print("Distance:");
30
31
32 void loop()
33 {
34   // Send ultrasonic pulse
35   digitalWrite(trigPin, LOW);
36   delayMicroseconds(5);
37   digitalWrite(trigPin, HIGH);
38   delayMicroseconds(10);
39
40
41

```

How the debugg
Add breakpoints b
numbers. Hover ov
paused to see the
buttons above to r
step one line at a

Serial Monitor

```

BUZZER ON
Distance: 59.77 cm
BUZZER ON
Distance: 60.01 cm
BUZZER ON
Distance: 60.01 cm
BUZZER ON

```

4. Learning outcomes (What I have learnt):

1. Ultrasonic sensors and its properties.
2. Interfacing of ultrasonic sensors with Arduino Uno and LCD.
3. Connecting ultrasonic sensor ,Arduino Uno board and LCD in a circuit.